PHYSICS ONLY

9 ELECTRIC CHARGE

Static electricity is the result of an imbalance of charge. It can be very useful, but it can also be extremely dangerous. In this chapter, you will learn about some of the uses and problems related to static electricity.

The photograph below shows a spectacular event caused by the build up of static electricity. Flashes of lightning are seen when thunderclouds discharge their electricity. The discharging currents can be as large as 20 000 A and typically take place in just 0.1 s. Such large currents cause the air to heat up to temperatures of approximately 30 000 °C. At such high temperatures, the air immediately around the flash of lightning expands at supersonic speeds (faster than the speed of sound), causing thunder. As the sound waves travel outwards, they slow down and interact with the air and ground along their paths. This causes the continuous low rumbling sounds we hear during a thunderstorm.



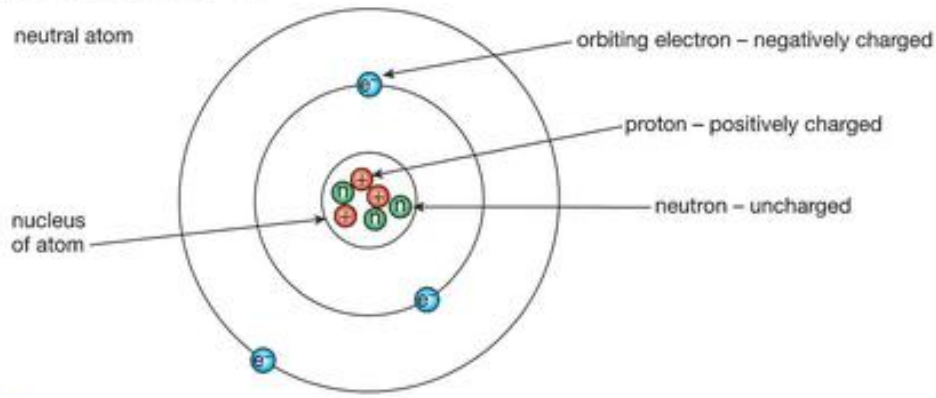
▲ Figure 9.1 Lightning strikes the surface of the Earth 100 times each second. That is 8 million times a day.

LEARNING OBJECTIVES

- Identify common materials which are electrical conductors or insulators, including metals and plastics
- Practical: investigate how insulating materials can be charged by friction
- Explain how positive and negative electrostatic charges are produced on materials by the loss and gain of electrons
- Know that there are forces of attraction between unlike charges and forces of repulsion between like charges
- Explain electrostatic **phenomena** in terms of the movement of electrons
- Explain the potential dangers of electrostatic charges, e.g. when fuelling aircraft and tankers
- Explain some uses of electrostatic charges, e.g. in photocopiers and inkjet printers

CHARGES WITHIN AN ATOM

All atoms contain small particles called protons, neutrons and electrons. The protons are found in the centre or nucleus of the atom and carry a **relative charge** of +1. The neutrons are also in the nucleus of the atom but carry no charge. The electrons travel around the nucleus in orbits. The electrons carry a relative charge of –1.



▲ Figure 9.2 A neutral atom has the same number of negative electrons and positive protons (not to scale).

Normally the number of protons in the nucleus is equal to the number of orbiting electrons. The atom therefore has no overall charge. It is neutral. If an atom gains extra electrons, it is then **negatively charged**. If an atom loses electrons, it becomes positively charged. An atom that becomes charged by gaining or losing electrons is called an ion.

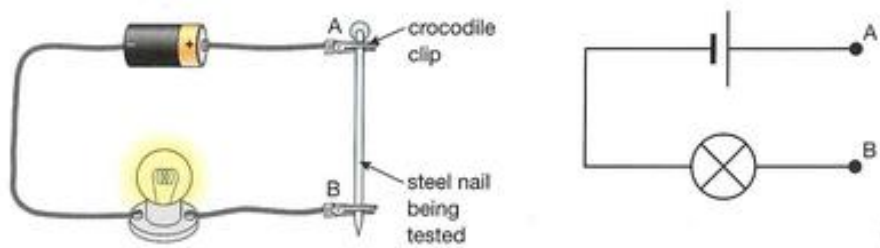
CHARGING MATERIALS BY FRICTION

KEY POINT

Some materials (such as metals) are good electrical conductors. Other materials that do not allow electricity to flow through them easily are known as insulators.

It is possible to charge some objects simply by rubbing them together. But these objects must be made from different materials and these materials must be **electrical insulators**.

We can test which materials are conductors and which are insulators using the circuit in Figure 9.3.



▲ Figure 9.3 Testing circuit

If we place a conductor between A and B the bulb will shine. If we place an insulator between A and B the bulb will not shine.

Materials such as graphite and metals such as copper, tin and gold allow electricity to flow through them easily. They are electrical conductors.

Materials such as plastic, rubber, glass and wood do not allow electricity to flow through them easily. They are electrical insulators.

REMINDER

It is important to remember that the rubbing action does not produce or create charge. It simply separates charge – that is, it transfers some electrons from one object to another.

ACTIVITY 1

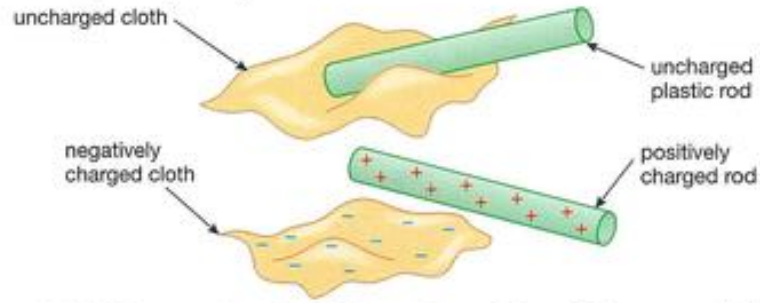
▼ PRACTICAL: INVESTIGATE HOW INSULATORS CAN BECOME ELECTRICALLY CHARGED BY FRICTION

We can show how insulating materials can be charged by friction using an uncharged plastic rod and an uncharged cloth. First we have to check they are not already charged. We can do this by placing the rod and the cloth in turn, close to some very small pieces of paper. If the cloth or the rod is charged, some of the paper will be attracted. If they are uncharged there will be no attraction (see page 88). Now we rub the rod and the cloth together for a few seconds and then retest them. Because they are made from different insulating materials they will be charged.

If we repeat this experiment using objects that are not insulators, or are made from the same material, they will not **attract** the small pieces of paper. That is, they remain uncharged.

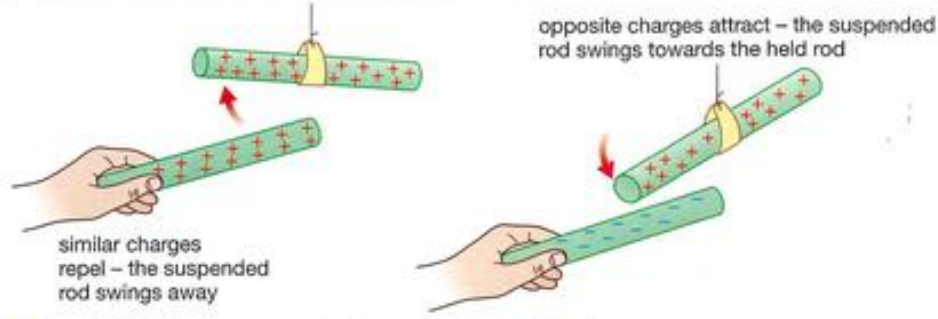
EXPLANATION

When the uncharged plastic rod is rubbed with an uncharged cloth, electrons from the atoms of the rod move onto the cloth. There is now an unequal number of positive and negative charges on each. The rod has lost electrons and so is positively charged. The cloth has gained electrons and so is negatively charged.



▲ Figure 9.4 Rubbing an uncharged rod with an uncharged piece of cloth can result in them both becoming charged.

FORCES BETWEEN CHARGES



▲ Figure 9.5 Similar charges repel and opposite charges attract.

Charged objects can exert forces on other charged objects without being in contact with them. If the charges are similar, the objects **repel** each other. If the objects are oppositely charged they attract each other.

KEY POINT

Static electricity is a build up of charge that does not move. The movement of charge is current electricity.

EXTENSION WORK

Although you will not be tested on the Van de Graaff generator it can be used to show dramatically the repulsion between similar charges.

Figure 9.6 shows what can happen if a person is charged with static electricity. The girl has her hands on a Van de Graaff generator. When it is turned on, charges flow onto the large metal dome. Some of the charges flow over her hands and onto all parts of her body, including her hair. Each strand (piece) of hair has the same type of charge as its neighbour. There are **repulsive forces** between the strands, so the strands repel each other. These forces cause her hair to stand on end.

For this demonstration to work, the girl must stand on an insulator to prevent any of the charges she is receiving from the generator from escaping into the floor. At the end of the demonstration the girl steps off the insulator – the charges can now escape and her hair falls. When a path is provided for charges to escape it is called earthing.



▲ Figure 9.6 A build-up of static electricity means that each hair on the girl's head has the same charge, and they repel each other.

FORCES BETWEEN CHARGED AND UNCHARGED OBJECTS

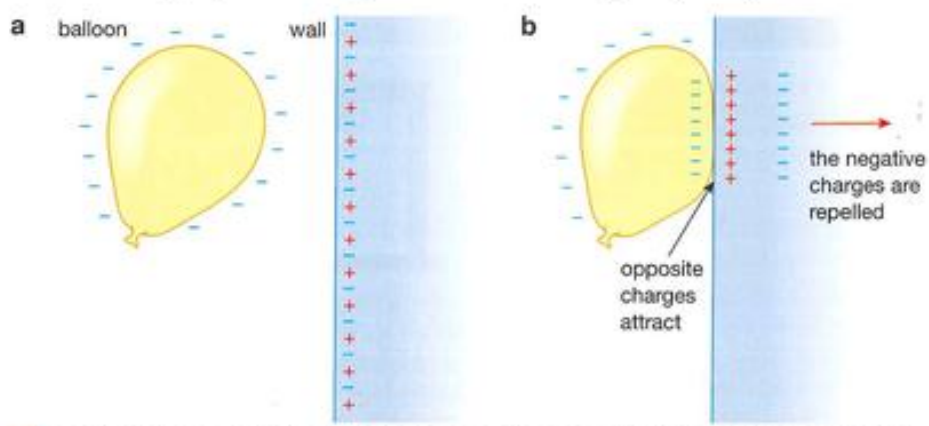
It is possible for a charged object to attract something that is uncharged. The balloon experiment demonstrates this.

If you charge a balloon by rubbing it against your jumper or your hair and then hold the balloon against a wall you will probably find that the balloon sticks to the wall. There is an attraction between the charged balloon and the uncharged wall.

EXPLANATION

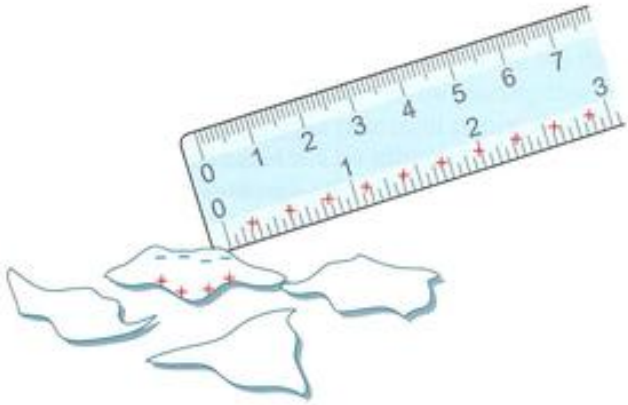
After the balloon has been charged with static electricity, but before it is brought close to the wall, the charges will be distributed as shown in Figure 9.7a. The balloon is charged (we have assumed negatively charged) and the wall is uncharged – that is, it has equal numbers of positive and negative charges.

As the negatively charged balloon is brought closer to the wall some of the negative electrons are repelled from the surface of the wall. This gives the surface of the wall a slight positive charge that attracts the negatively charged balloon.



▲ Figure 9.7 a The negatively charged balloon approaches the neutrally charged surface of the wall. b The positive charge on the surface of the wall attracts the oppositely charged balloon.

You can try a similar experiment using a plastic ruler and some small pieces of paper. Rub the plastic ruler on the sleeve of your jumper. The ruler will become charged. (We have assumed that the ruler has become positively charged.) If it is held close to some small, uncharged pieces of paper some electrons within the paper will be attracted to the edges closest to the ruler. There will be an attraction between these negative parts of the paper and the positive ruler.



▲ Figure 9.8 The charged ruler induces a charge in the paper, and the two attract.

KEY POINT

The charges that appear on the pieces of paper are called **induced** charges. When the ruler is removed the charges redistribute (rearrange) themselves so that the pieces of paper are once again uncharged.

EXTENSION WORK

Take a plastic ruler or rod and rub it against a jumper to charge it with static electricity. Turn on a water tap so that the water flows from the tap as slowly as possible but as a continuous flow (not as a series of drops). Hold the charged rod or ruler close to the stream of water but not in it. What happens to the water? Can you explain what is happening?

EXTENSION WORK

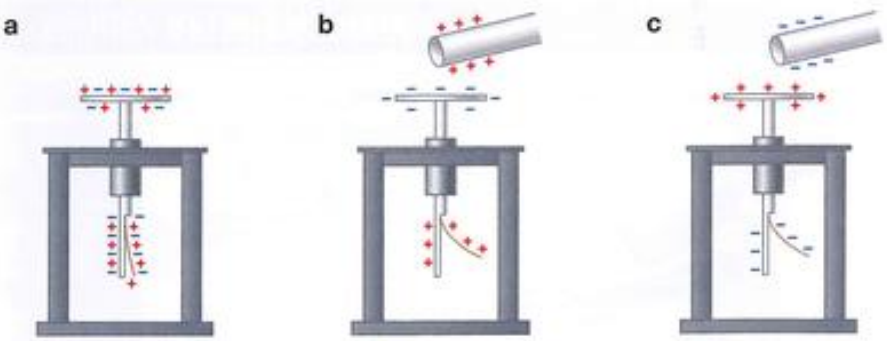
The gold leaf electroscope



▲ Figure 9.9 A modern gold leaf electroscope

The gold leaf electroscope is a very useful instrument for detecting charge. It is not mentioned in your examination specification but if you can understand how it works you will have understood the most important properties of static electricity.

The metal plate at the top of the electroscope is connected to a metal rod inside the case. The rod has a small thin piece of metal (called a 'leaf') attached to it. This is usually made of gold, as gold can be made into extremely thin sheets. If an electrically charged object is brought close to the plate at the top, the gold leaf will move.

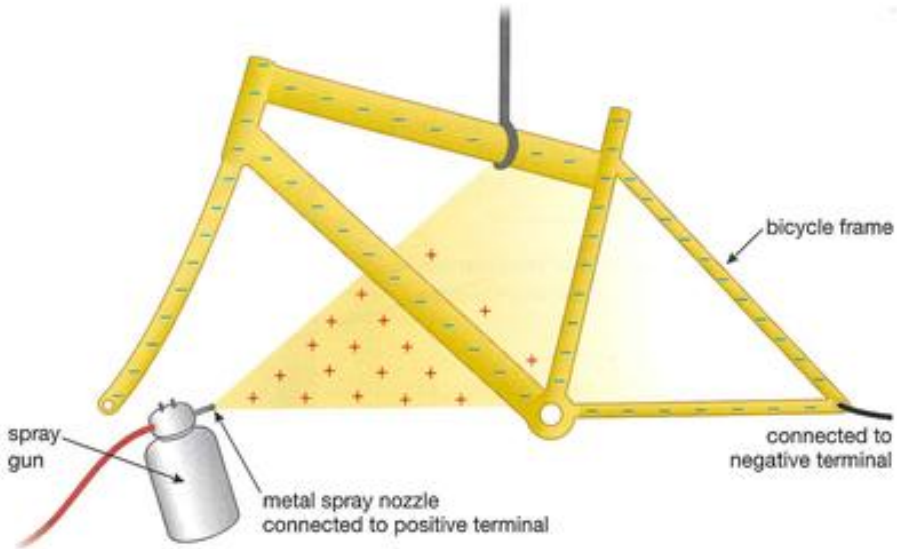


▲ Figure 9.10 **a** If there are no electrically charged objects held close to the plate the positive and negative charges are evenly spread throughout the electroscope and the gold leaf is not deflected. **b** If a positively charged object is brought close it attracts negative charges upwards onto the metal plate. This leaves both the metal rod and the gold leaf positively charged. They repel each other, so the leaf rises. **c** If a negatively charged object is brought close it repels the negatively charged electrons on the plate pushing them down onto the gold leaf and the bottom of the rod. These like charges repel and the leaf again moves upwards.

USES OF STATIC ELECTRICITY

ELECTROSTATIC PAINT SPRAYING

Painting an awkwardly shaped object, such as a bicycle frame, with a spray gun can take a long time and use a lot of paint. Using electrostatic spraying can make the process much more efficient.



▲ Figure 9.11 The positive paint is attracted to all parts of the negatively charged object.

As the drops of paint emerge from the spray gun, they are charged. As the drops all carry the same charge they repel and spread out forming a thin spray. The metal bicycle frame has a wire attached to an electrical supply giving the frame the opposite charge. The paint drops are therefore attracted to the surface of the frame. There is the added benefit that paint is attracted into places, such as corners, that would normally be hard to reach.

INKJET PRINTERS



▲ Figure 9.12 An inkjet printer

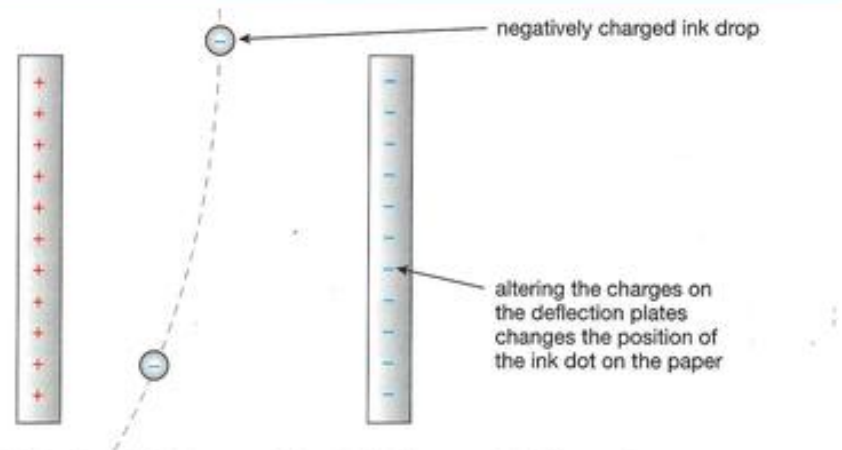
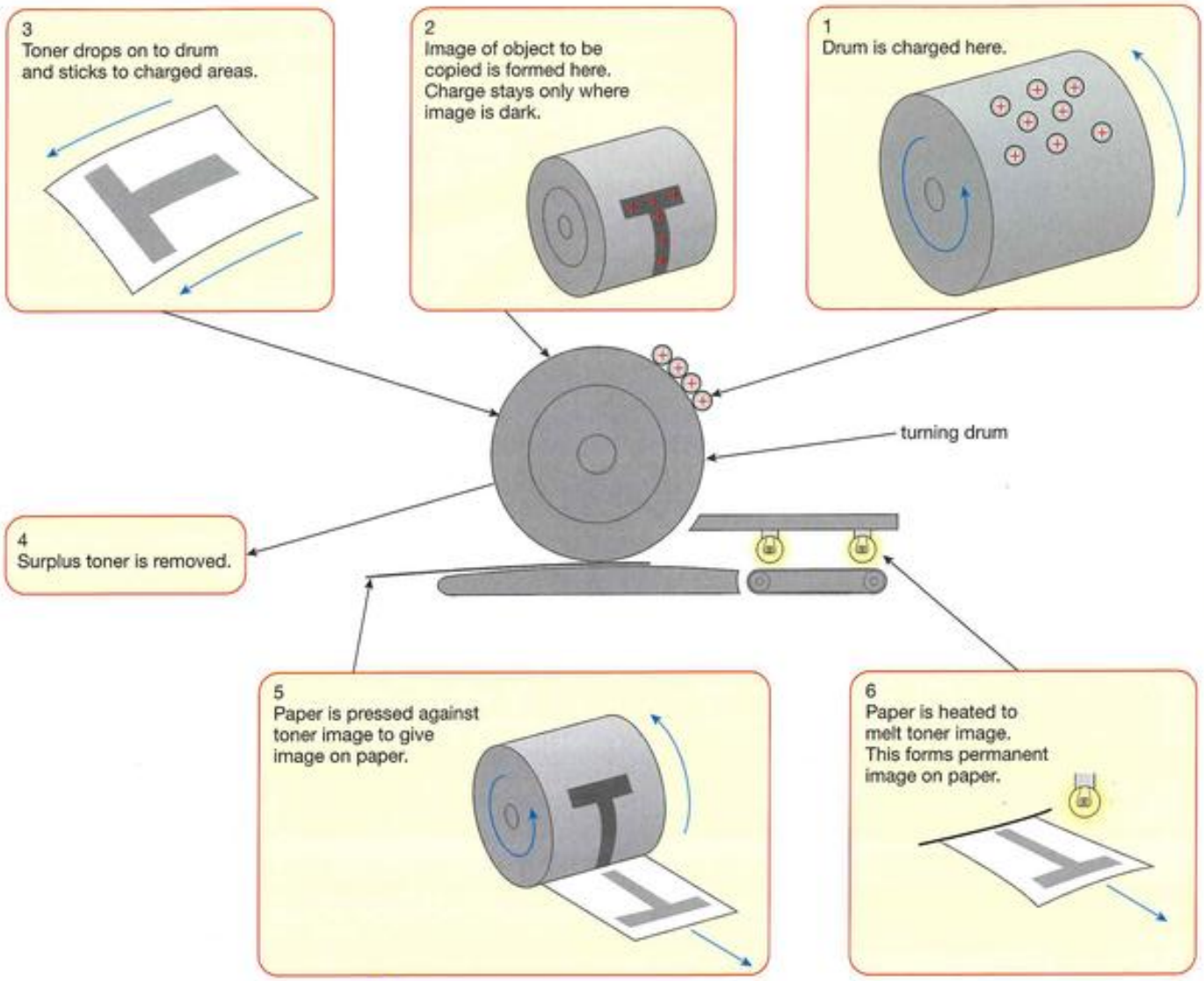


Figure 9.13 The charged ink drops are deflected into the correct position on the paper.

Many modern **inkjet printers** use inkjets to direct a fine jet (stream) of ink drops onto paper. They do this by using electrostatic forces. Each spot of ink is given a charge so that as it falls between a pair of deflecting plates, electrostatic forces direct it to the correct position. The charges on the plates change hundreds of times each second so that each drop falls in a different position, forming pictures and words on the paper as required.

PHOTOCOPIERS



▲ Figure 9.14 Static electricity is used in photocopiers.

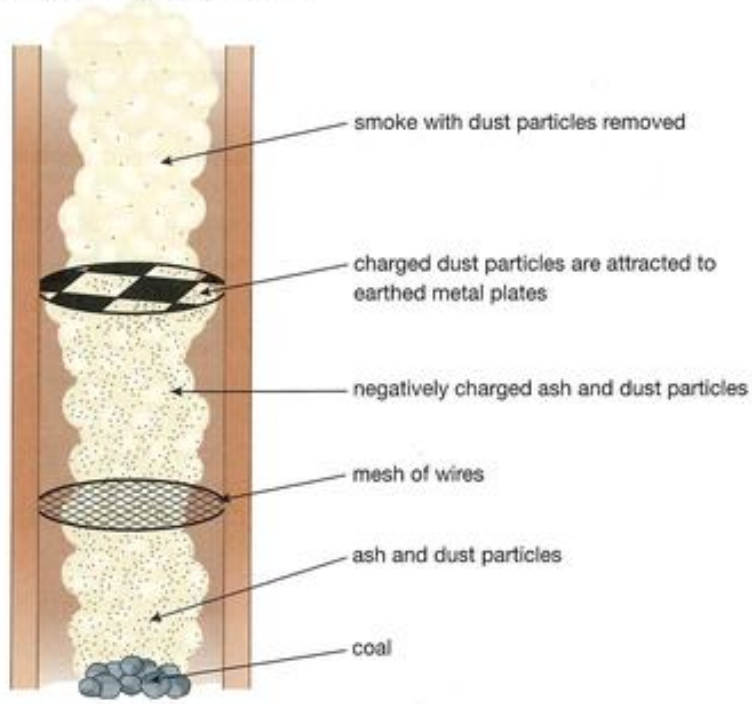


▲ Figure 9.15 A photocopier

Positive charges are sprayed onto a turning drum whose surface is covered with a metal called selenium. A bright light is shone onto the sheet of paper to be copied. The white parts of the paper reflect light onto the drum; the dark or printed parts do not. In those places where light is reflected onto the drum the selenium loses its charge but where no light is reflected onto the drum the charge remains. A negatively charged carbon powder called toner is blown across the drum and sticks to just those parts of the drum that are charged. A sheet of paper is now pressed against the drum and picks up the pattern of the carbon powder. The powder is then fixed in place by a heater.

ELECTROSTATIC PRECIPITATORS

Many heavy industrial plants, such as steel-making furnaces and coal-fired power stations, produce large quantities of smoke. This smoke carries small particles of ash and dust into the environment, causing health problems and damage to buildings. One way of removing these pollutants from the smoke is to use electrostatic precipitators.



▲ Figure 9.16 Electrostatic precipitators help to cut down the amount of pollution released into the atmosphere.

As the smoke initially rises up the chimney, it passes through a mesh of wires that are highly charged. (The wires are at a voltage of approximately −50 000 V.) As they pass through the mesh, the ash and dust particles become negatively charged. Higher up the chimney, these charged particles are attracted by and stick to, large metal earthed plates. The cleaner smoke is then released into the atmosphere. When the earthed plates are completely covered with dust and ash, they are tapped hard. The dust and ash fall into collection boxes, which are later emptied.

In a large coal-fired power station, 50–60 tonnes of dust and ash may be removed from the smoke each hour!



▲ Figure 9.17 Earthing during refuelling

PROBLEMS WITH STATIC ELECTRICITY

In some situations the presence of static electricity can be a disadvantage.

As aircraft fly through the air, friction causes them to become charged with static electricity. After an aircraft has landed there is the possibility of charges escaping to earth as a spark or flash of electricity. If this takes place during refuelling, it could cause an explosion. The solution to this problem is to earth the plane with a conductor as soon as it lands and before refuelling begins, allowing the charge that has built up to flow to earth. Fuel tankers that transport fuel on roads must also be earthed before any fuel is transferred, to prevent sparks causing a fire or an explosion.

Sometimes after a long car journey on a dry day we can become charged with static electricity and when we step from the car we might receive a small electric shock.

Our clothing can become charged with static electricity under certain circumstances. When we remove the clothes there is the possibility of receiving a small electric shock as the charges escape to earth.

CHAPTER QUESTIONS

More questions on domestic electricity can be found at the end of Unit 2 on page 93.

SKILLS CRITICAL THINKING

7

1 a What charge is carried by each of these particles?
   i a proton
   ii an electron
   iii a neutron
   b Where inside an atom are each of the three particles mentioned in part a found?
   c How many protons are there in a neutral atom compared to the number of electrons?

6

   d What do we call an atom that has become charged by gaining or losing electrons?

SKILLS INTERPRETATION

   e Describe with diagrams how two objects can be charged by friction (rubbing).

SKILLS REASONING

7

2 Explain the following.
   a A crackling sound can sometimes be heard when removing a shirt.
   b Sometimes after a journey in a car you can get a small electric shock when you touch the handle of the door.
   c A plastic comb is able to attract small pieces of paper immediately after it has been used.
   d After landing, aircraft are always 'earthed' before being refuelled.

SKILLS ADAPTIVE LEARNING

SKILLS REASONING

3 a In a photocopier, why does toner powder stick to some places on the selenium-covered drum but not to others?
   b Explain why ash and dust particles are attracted towards the earthed metal plates of an electrostatic precipitator after they have passed through a highly negatively charged mesh of wires.

HINT

Read again about the balloon experiment.

SKILLS CRITICAL THINKING

4 Lightning is caused by clouds discharging their static electricity.
   a Find out:
      i how the clouds become charged
      ii how a lightning conductor works.
   b Suggest two places which might be
      i unsafe during a thunderstorm
      ii safe during a thunderstorm.

SKILLS REASONING

5 Computer chips can be damaged by static electricity if a spark jumps between a worker and a computer. Suggest how workers who build and repair computers avoid this problem.

HINT

See section 'Problems with static electricity'.

END OF PHYSICS ONLY